

SYSTEM DOCUMENTATION

AI-Powered Coffee Experience

How Raft Coffee built a mood-based ordering platform with an intelligent chatbot for Lahore, Pakistan.



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01

OVERVIEW

A coffee shop that reads your mood.

Raft Coffee is a specialty shop in Lahore that recommends beverages based on emotional state — powered by RAG and LLMs.

System Overview

RAFT COFFEE IS A MOOD-BASED SPECIALTY COFFEE SHOP WEB APPLICATION, COMBINING A MODERN E-COMMERCE FRONTEND WITH AN AI-POWERED CHATBOT BACKEND.

The system helps customers discover coffee blends tailored to their emotional state through an interactive chat interface, guided by a goose mascot chatbot trained on Raft's full menu and brand knowledge.

The application features mood-matched beverages, shopping cart functionality, product customization, and an admin dashboard — all built with a modular, scalable architecture.



FRONTEND

React 18 + TypeScript SPA, built with Vite and styled with Tailwind CSS and Radix UI components.

BACKEND

FastAPI (Python) with LangChain, Groq LLM API, and FAISS vector search for RAG-powered responses.

AUTHENTICATION

Supabase powers secure admin login and session management.

DATA FLOW

- 1 → User sends message via chat
- 2 → POST /api/chat endpoint
- 3 → FAISS retrieves relevant context
- 4 → LangChain constructs prompt
- 5 → Groq LLM generates response
- 6 → Response rendered in chat UI

02

THE CHALLENGE

Coffee ordering is emotional, not just transactional.

Most coffee websites list products. Raft wanted to understand how you feel first — then recommend the perfect cup.

The Problem We Solved

RAFT'S TEAM FACED A PRODUCT VISION — THEY WANTED CUSTOMERS TO FIND THE RIGHT COFFEE FOR THEIR MOOD, NOT JUST BROWSE A STATIC MENU.

01

GENERIC MENUS

Standard coffee menus give no emotional context. A tired customer and an anxious one need completely different beverages.

02

NO PERSONALIZATION

Static e-commerce cannot adapt to user state. Every customer gets the same experience regardless of their mood or context.

03

KNOWLEDGE GAPS

Customers unfamiliar with specialty coffee struggle to choose. There was no interactive guide to help them navigate the menu.



The solution required building a conversational AI that could read between the lines of a customer's message — detecting mood signals and mapping them to Raft's curated menu of signature, classic, and seasonal blends.

Using RAG (Retrieval-Augmented Generation), the chatbot draws on a rich knowledge base of Raft's menu, brand story, and coffee expertise to give accurate, on-brand responses — not generic AI outputs.

03

AI CHATBOT

A system that understands how you feel.

*RAG + LangChain + Groq — three layers working
together to deliver instant, contextual coffee
recommendations.*

Raft Coffee · System Documentation

AI Chatbot Architecture

AI Chatbot Architecture

THE CHATBOT USES RETRIEVAL-AUGMENTED GENERATION TO COMBINE DOCUMENT SEARCH WITH LLM GENERATION — GIVING RESPONSES GROUNDED IN REAL RAFT COFFEE KNOWLEDGE.

FR-1.1

MESSAGE ENDPOINT

User messages arrive via POST /api/chat. FastAPI validates input with Pydantic before passing to the chatbot service.

FR-1.2

VECTOR RETRIEVAL

FAISS performs sub-millisecond similarity search over pre-computed sentence-transformer embeddings to find relevant context.

FR-1.3

LANGCHAIN ORCHESTRATION

Context + user message + system prompt are assembled into a LangChain chain that invokes the Groq LLM asynchronously.

FR-1.4

GRACEFUL ERRORS

All API failures are caught and surfaced as friendly messages — the system degrades gracefully even without AI.

PERFORMANCE TARGETS

30–60s

MODEL PRE-LOAD ON STARTUP

<2s

CHAT RESPONSE AFTER LOAD

<1ms

FAISS VECTOR SEARCH



E-Commerce & Cart System

THE MENU SYSTEM ORGANIZES COFFEES BY MOOD CATEGORIES — SIGNATURE, CLASSIC, AND SEASONAL — WITH FULL CART AND CHECKOUT FUNCTIONALITY.

FR-2.1

MOOD MENU

Items carry mood tags (e.g. focused, relaxed, energized) that inform both browsing and chatbot recommendations.

FR-2.2

CATEGORY FILTER

Menu filtered dynamically by signature, classic, and seasonal categories. Filter state is managed in component state.

FR-2.3

CUSTOMIZATION

Each item supports size (small, regular, large) and flavor profile selection before adding to cart.

FR-3.1

ADD TO CART

CartContext provides global add/remove/update operations. Cart state persists in component state during session.

FR-3.3

LIVE TOTALS

Item count and total price recalculate reactively. Toast notifications confirm cart actions.

FR-4.1

ADMIN PANEL

Supabase auth protects admin routes. Session management with automatic logout on expiry.



Technology Stack

A MODERN, MODULAR STACK DESIGNED FOR INDEPENDENT SCALING OF FRONTEND AND BACKEND WITH STRONG TYPE SAFETY THROUGHOUT.

FRONTEND

React 18 TypeScript Vite
Tailwind CSS Radix UI React Router
Lucide Icons

BACKEND

FastAPI Python 3.8+ LangChain
Uvicorn Pydantic python-dotenv

AI & SEARCH

Groq API (Llama) FAISS (CPU)
Sentence Transformers all-MiniLM-L6-v2

INFRASTRUCTURE

Supabase Auth Vercel / Netlify
Railway / Render

DESIGN CONSTRAINTS

- DC-1** · Frontend must use React 18+ with TypeScript
- DC-2** · Backend must use FastAPI + Python 3.8+
- DC-3** · Styling must use Tailwind CSS
- DC-4** · UI components must use Radix UI
- DC-5** · LLM must use Groq API

DEPLOYMENT

Frontend → Static hosting (Vercel / Netlify)
Backend → Cloud platform (Railway / Render)
Auth → Supabase managed service
CI/CD → GitHub Actions

File Structure

BACKEND

backend/app/

main.py — FastAPI entry point

routes/

chat.py — POST /api/chat

services/

chatbot.py — LangChain logic

llm.py — Groq config

retrieval.py — FAISS search

core/

config.py — env vars

prompts.py — system prompt

FRONTEND

frontend/src/

App.tsx — router + auth wrap

pages/

Index.tsx — homepage

Menu.tsx — menu + filter

components/

ChatBubble.tsx

ChatWindow.tsx

contexts/

CartContext.tsx

AdminAuthContext.tsx

data/

menu.ts — static menu items

Results

RAFT COFFEE DELIVERED A FULL-STACK MOOD-BASED ORDERING PLATFORM — THE FIRST OF ITS KIND IN LAHORE — COMBINING CONVERSATIONAL AI WITH A POLISHED E-COMMERCE EXPERIENCE.

MENU CATEGORIES 3 Signature · Classic · Seasonal	AI RESPONSE TIME < 2s After model pre-load
TECH STACK LAYERS 4 Frontend · Backend · AI · Auth	EMBEDDING MODEL MiniLM-L6 all-MiniLM-L6-v2 · FAISS CPU
FUNCTIONAL REQUIREMENTS 16 Across 4 system modules	PLATFORM TARGET Lahore, PK Specialty coffee market

PROJECT SUMMARY

Raft Coffee demonstrates that specialty retail can move beyond transactional interfaces — using AI to create genuinely personalized experiences that match product to person at the moment of decision.



Legal

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